

COMPLETE MATHS

ब्रह्मास्त्र

FORMULA BOOK

- CONCEPTS
- CLASS NOTES
- SHORT TRICKS
- SOLVED EXAMPLES
- CALCULATION TRICKS



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DEFENCE, BANKING,
& ALL GOVT. EXAMS**



ADITYA RANJAN SIR
(EXCISE INSPECTOR)

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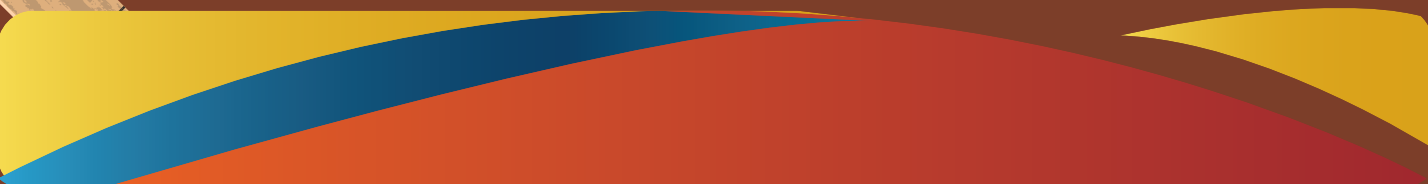
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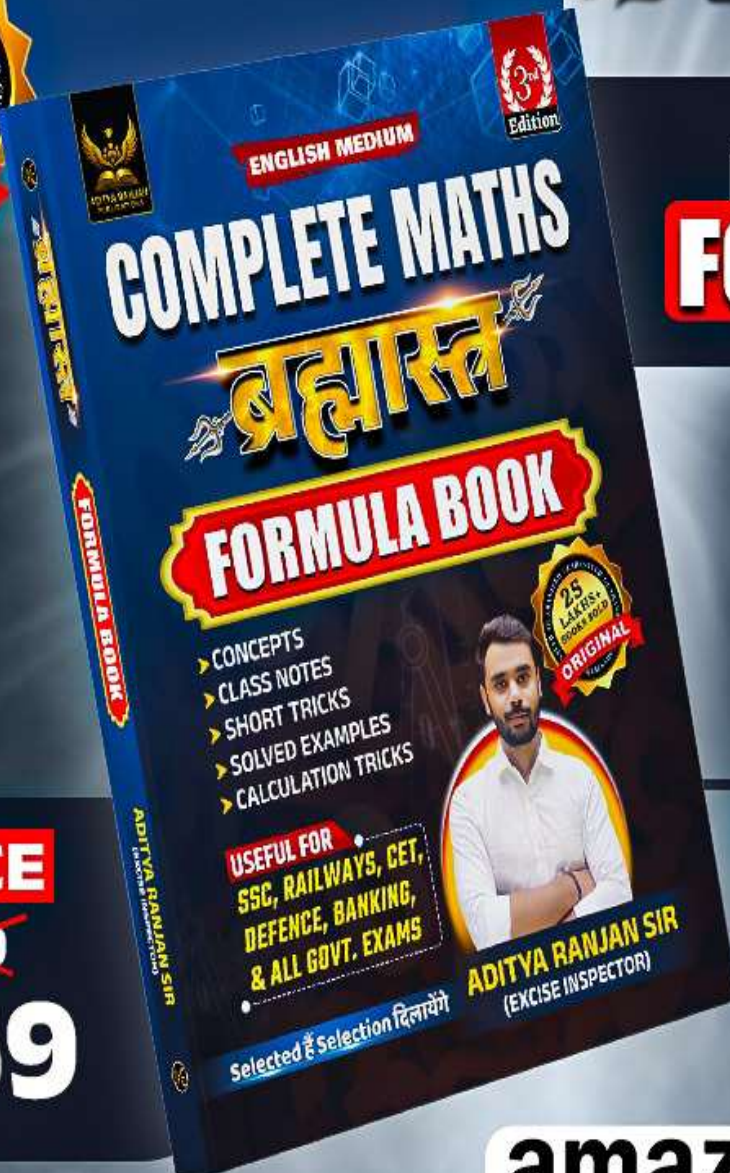


3rd EDITION

ENGLISH MEDIUM



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GEOMETRY

Lines & Angle

Triangle

Quadrilateral

Polygon

Circle

LINES & ANGLE

Point

A point is a circle of zero radius which determines a location. It is usually denoted by a capital letter.

Types of Points

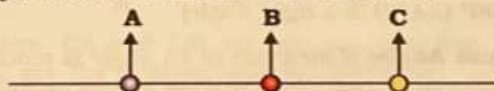
Collinear Points

Non-Collinear Points

Intersecting Points

(a) **Collinear Points:** If three or more points situated on a straight line, these points are called collinear points.

Example : Points A, B, and C are collinear.



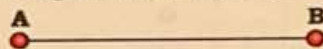
(b) **Non-collinear Points:** If three or more points are not situated on a straight line, these points are called non-collinear points.

(c) **Intersecting Points:** When two or more lines cross or meet at a common point, that point is known as intersecting point.

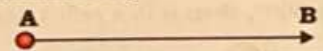
Line

A line is made up of an infinite number of points and it has only length i.e., it does not have any thickness (or width). A line is endless so, it can be extended in both directions.

Line Segment: A line segment has two end points; generally line segment is called line.



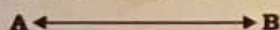
Ray: A ray extends indefinitely in one direction from any given point is called ray.



Name	Line	Line segment	Ray
Symbol	$\longleftrightarrow AB$	$\overline{AB}$	$\rightarrow AB$
End Points	No end points	Two end points	One end points

Types of Lines

(a) **Straight line:** A line which does not change its direction at any point is called a straight line.

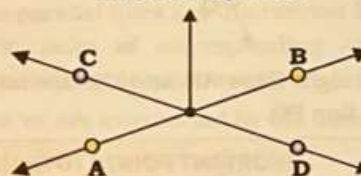


(b) **Curve line:** A line which changes its direction is called a curved line.

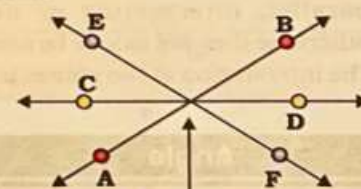


(c) **Intersecting Lines:** If two or more lines intersect each other, then they are called intersecting lines. In the figure AB and CD are intersecting lines. Two lines can intersect maximum at one point.

Intersecting Point

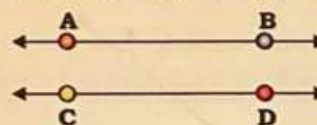


(d) **Concurrent Lines:** If three or more lines pass through a point, then they are called concurrent lines and the point through which these all lines pass is called point of concurrent.



Concurrent point

(e) **Parallel Lines:** Two straight lines are parallel if they lie in the same plane and do not intersect even if they produced infinitely. Perpendicular distances between two parallel lines are always same at all places.

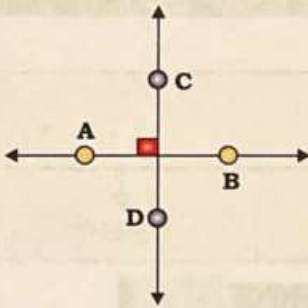


In the figure AB and CD are parallel lines.

Symbol for parallel lines is $\parallel$.

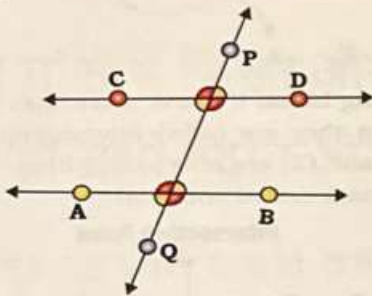
Hence, $AB \parallel CD$.

- (f) **Perpendicular Lines** : If two lines intersect each other at right angles, then two lines are called perpendicular lines. In the following figure AB and CD are perpendicular lines.



Symbolically it is represented as $AB \perp CD$ or we can also say that $CD \perp AB$.

- (g) **Transversal Lines** : A line which intersects two or more given lines at distinct points is called a transversal of the given lines.



In figure straight lines AB and CD are intersected by a transversal line PQ.

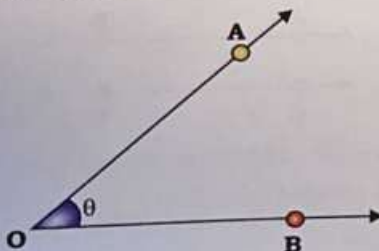
IMPORTANT POINTS TO REMEMBER

- There is one and only one line passing through two distinct points.
- Two or more lines are said to be coplanar if they lie in the same plane and can be parallel, intersecting or overlapping, otherwise they are said to be non-coplanar.
- The intersection of two planes is a line.

Angle

An angle is the union of two non-collinear line with a common initial point. The two line forming an angle are called arms of the angle and the common initial point is called the vertex of the angle.

- The angle AOB is denoted by $\angle AOB$, is formed by line OA and OB and point O is the "vertex" of the angle.

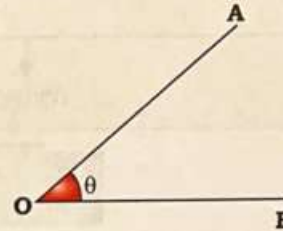


(I)

Types of Angle

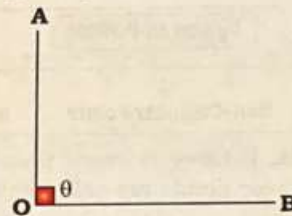
Acute angle Right angle Obtuse angle Straight angle Reflex angle Complete angle

Acute Angle: If the measure of an angle is less than 90° , it is an acute angle.



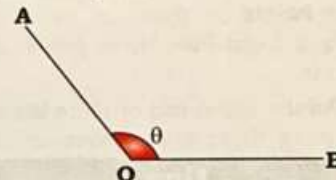
$0^\circ < \theta < 90^\circ$ ($\angle AOB$ is an acute angle)

Right Angle : If measure of an angle is equal to 90° , then it is a right angle.



$\theta = 90^\circ$ ($\angle AOB$ is a right angle)

Obtuse Angle: If measure of an angle is more than 90° but less than 180° , then it is an obtuse angle.



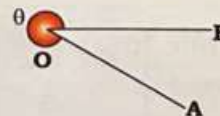
$90^\circ < \theta < 180^\circ$ ($\angle AOB$ is an obtuse angle)

Straight Angle: If measure of an angle is equal to 180° , then it is a straight angle.



$\theta = 180^\circ$ ($\angle AOB$ is a straight angle)

Reflex Angle: If measure of an angle is more than 180° but less than 360° , then it is a reflex angle.



$180^\circ < \theta < 360^\circ$ ($\angle AOB$ is a reflex angle)

Complete angle: If measure of an angle is 360° then it is a complete angle.



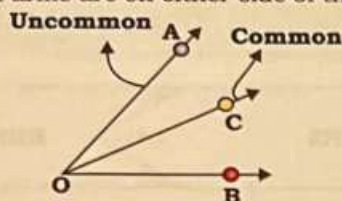
(II)

Pair of Angles

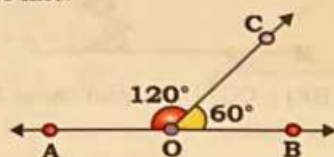
Adjacent Angle **Linear Angle** **Complementary Angle** **Supplementary Angle**

Adjacent Angles: Two angles are called adjacent angles if:

- They have the same vertex,
- They have a common arm,
- Uncommon arms are on either side of the common arm.



- In the figure, $\angle AOC$ and $\angle BOC$ have a common vertex O. Also, they have a common arm OC and the distinct arms OA and OB, lie on the either side of common arm OC.
- **Linear Pair of Angles:** Two adjacent angles are said to form a linear pair of angles, if their uncommon arms are two opposite line.



- In figure, OA and OB are two opposite line $\angle AOC$ & $\angle BOC$ are the adjacent angles. Therefore, $\angle AOC$ and $\angle BOC$ form a linear pair.
- If a line stand on another line, the sum of the adjacent angles so formed is 180° .

Angle	Complementary	Supplementary
Definition	If sum of two angles is equal to 90° , then the two angles are called complementary angles. • $\angle BAD$ and $\angle DAC$ are complementary angles, if $x^\circ + y^\circ = 90^\circ$	If sum of two angles is equal to 180° , then the two angles are called supplementary angle. • $\angle BAC$ and $\angle DAC$ are supplementary angles, if $x^\circ + y^\circ = 180^\circ$
Representation		

Ex. The measure of an angle for which the measure of the supplement is four times the measure of the complement is:



HINTS Let Angle = x .

So, Complementary angle = $90^\circ - x$

Supplementary angle = $180^\circ - x$

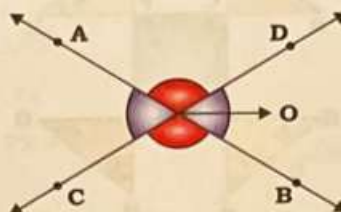
ATQ,

$4 \times \text{Complementary angle} = \text{Supplementary angle}$

$\Rightarrow 4 \times (90^\circ - x) = 180^\circ - x \Rightarrow x = 60^\circ$

(III) Vertically Opposite Angles

If arm of two angles form two pairs of opposite rays, then the two angles are called as vertically opposite angles.



- In other words, when two lines intersect, two pairs of vertically opposite angles are formed. Each pair of vertically opposite angle is equal.
- In the figure, two lines AB and CD intersect at O. We find that $\angle AOC$ and $\angle BOD$ are vertically opposite angles
 $\Rightarrow \angle AOC = \angle BOD$
Similarly, $\angle BOC$ and $\angle AOD$ are vertically opposite angles.
 $\Rightarrow \angle BOC = \angle AOD$

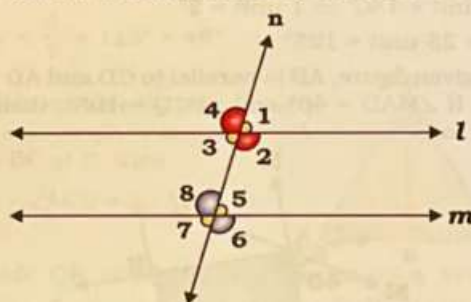
(IV)

Angles made by Transversal Lines

Corresponding Angle **Alternate Angle** **Consecutive Interior Angle**

When two parallel lines are intersected by a transversal. They form pairs of corresponding angles, Alternate angle, Consecutive Interior angles.

Lines 'l' and 'm' are intersected by the transversal 'n'. Then



Corresponding Angle

$\angle 1 = \angle 5$, $\angle 4 = \angle 8$, $\angle 3 = \angle 7$ and $\angle 2 = \angle 6$

Alternate Interior Angle

$\angle 3 = \angle 5$ and $\angle 2 = \angle 8$

Alternate Exterior Angle

$\angle 1 = \angle 7$ and $\angle 4 = \angle 6$

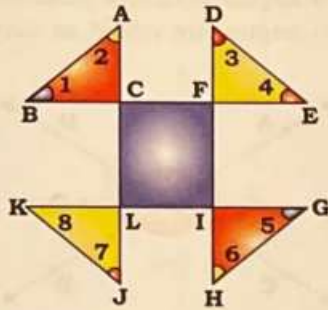
Consecutive Interior Angles

$\angle 2 + \angle 5 = 180^\circ$ & $\angle 3 + \angle 8 = 180^\circ$



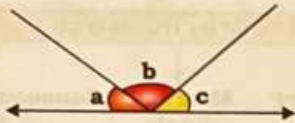
Conversely, if a transversal intersects two lines in such a way that a pair of alternate interior angles is equal, then the two lines are parallel.

Ex. Angles are shown in the given figure. What is the value of $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8$?



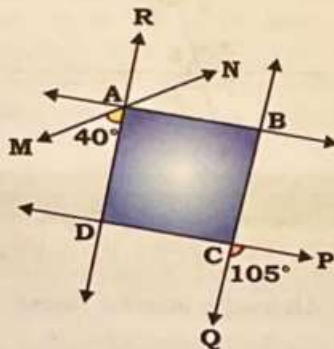
HINTS $\angle C = 180^\circ - (\angle 1 + \angle 2) \Rightarrow \angle F = 180^\circ - (\angle 3 + \angle 4)$
 $\Rightarrow \angle I = 180^\circ - (\angle 5 + \angle 6) \Rightarrow \angle L = 180^\circ - (\angle 7 + \angle 8)$
 CFIL is quadrilateral.
 $\therefore$ Sum of angles of CFIL is 360° .
 $\Rightarrow 720^\circ - (\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8) = 360^\circ$
 $\Rightarrow 360^\circ = (\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8)$

Ex. In the given figure, if $\frac{b}{a} = 5$ and $\frac{c}{a} = \frac{6}{5}$ then what is the value of b .



HINTS $b : a = (5 : 1) \times 5 = 25 : 5$
 $c : a = 6 : 5$
 $\therefore a : b : c = 5 : 25 : 6$
 $\angle a + \angle b + \angle c = 180^\circ$ (Straight line)
 $\Rightarrow 36 \text{ unit} = 180^\circ \Rightarrow 1 \text{ unit} = 5^\circ$
 $\therefore \angle b = 25 \text{ unit} = 125^\circ$

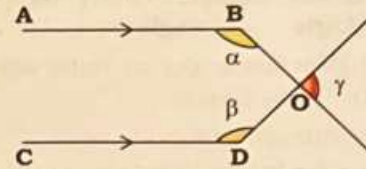
Ex. In the given figure, AB is parallel to CD and AD is parallel to BC. If $\angle MAD = 40^\circ$ and $\angle PCQ = 105^\circ$, then $\angle NAB$ is equals to?



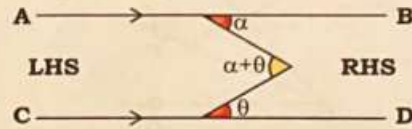
HINTS $\angle BCD = \angle PCQ = 105^\circ$ (Vertically opposite angle)
 $\angle CDA = 180^\circ - 105^\circ = 75^\circ$
 $\angle RAB = \angle CDA = 75^\circ$ (corresponding angles)
 $\angle RAN = \angle MAD = 40^\circ$ (Vertically opposite angle)
 $\therefore \angle NAB = \angle RAB - \angle RAN = 75^\circ - 40^\circ = 35^\circ$

Some Important Results

Ex If $AB \parallel CD$ then $\alpha + \beta + \gamma = 360^\circ$



Ex If $AB \parallel CD$, then sum of angle on left hand side is equal to sum of angle on right hand side.

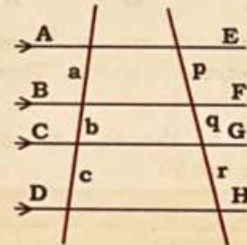


RHS = LHS
 $\Rightarrow \theta + \alpha = \alpha + \theta$

Ex If $L \parallel M$, $a + b + c = \alpha + \beta + \gamma$



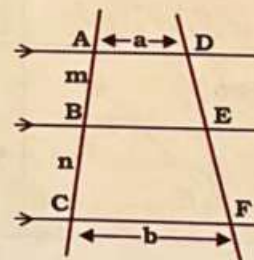
Ex When $AE \parallel BF \parallel CG \parallel DH$ and these lines are cut by two transversal lines.



Then, $a : b : c = p : q : r$

$$\frac{a}{a+b+c} = \frac{p}{p+q+r}$$

Ex When $AD \parallel BE \parallel CF$ and these lines are cut by two transversal lines.

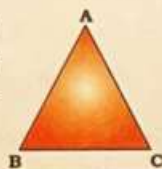


Then, $\frac{AB}{BC} = \frac{DE}{EF} = \frac{m}{n}$

$$BE = \frac{an + bm}{m + n}$$

TRIANGLE

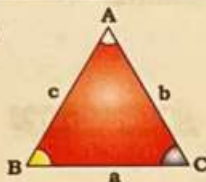
A triangle is a polygon with three sides and three angles. It is the fundamental shape in geometry formed by connecting three non-collinear points.



Fundamental Properties of Triangle

Ex. Sum of all three angles of a triangle is always 180°

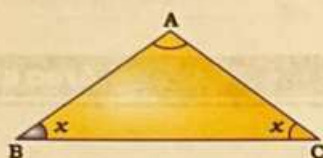
$$\angle A + \angle B + \angle C = 180^\circ$$



Ex. One of the angles of a triangle is 108° and the other two angles are equal. What is the measure of each of these equal angles?

HINTS We know,

$$\begin{aligned}\angle A + \angle B + \angle C &= 180^\circ \\ \Rightarrow 108^\circ + x + x &= 180^\circ \\ \Rightarrow 2x &= 72^\circ \Rightarrow x = 36^\circ\end{aligned}$$



(i) Angles opposite to the equal sides of a triangle are equal.

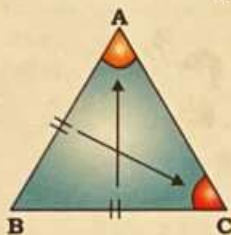
(ii) Sides opposite to the equal angles of a triangle are equal.

In $\triangle ABC$, If $AB = BC$, then

$$\angle A = \angle C$$

In $\triangle ABC$, If $\angle A = \angle C$, then

$$AB = BC$$

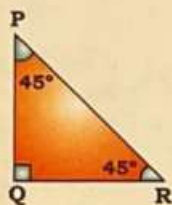


Ex. In $\triangle PQR$, $\angle Q = 90^\circ$, $PQ = 8$ cm and $\angle PRQ = 45^\circ$. Find the length of QR .

HINTS

Using property 2(ii),

$$QR = PQ = 8 \text{ cm}$$



(i) The angle opposite to the greater side is always greater than the angle opposite to the smaller side.

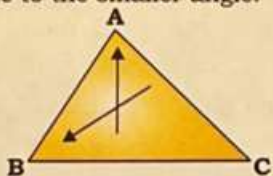
(ii) The side opposite to the greater angle is always greater than the side opposite to the smaller angle.

In $\triangle ABC$, If $BC > AC$, then

$$\angle A > \angle B$$

In $\triangle ABC$, If $\angle A > \angle C$, then

$$BC > AB$$



Ex. The ratio of the angles, $\angle P$, $\angle Q$ and $\angle R$ of a $\triangle PQR$ is $2 : 4 : 9$, then which of the following is true?

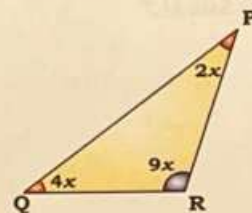
- $PQ > QR > RP$
- $PQ > RP > QR$
- $QR > RP > PQ$
- $PR > PQ > QR$

HINTS

Here, In $\triangle PQR$

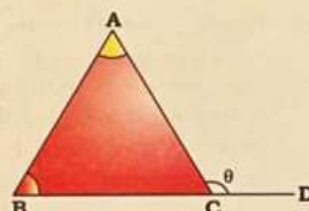
$$9x > 4x > 2x$$

$$\text{So, } PQ > PR > QR$$



If a side of triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.

$$\angle ACD = \angle CAB + \angle ABC$$



Ex. The side BC of $\triangle ABC$ is produced to D . If $\angle ACD = 112^\circ$ and $\angle B = \frac{3}{4} \angle A$, then find the measure of $\angle B$.

HINTS We know,

$$\text{Exterior angle } (\angle ACD)$$

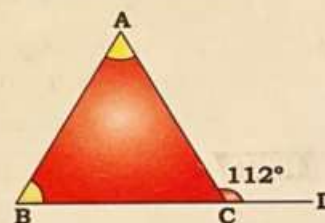
$$= \angle A + \angle B$$

$$\therefore \angle B = \frac{3}{4} \angle A$$

$$\Rightarrow \angle A = \frac{4}{3} \angle B \quad \dots (i)$$

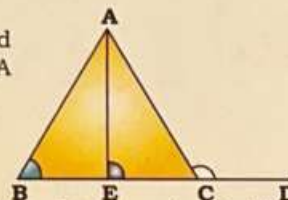
$$\therefore 112^\circ = \frac{4}{3} \angle B + \angle B$$

$$\Rightarrow \angle B = \frac{3}{7} \times 112^\circ = 48^\circ$$



In $\triangle ABC$, the side BC is produced to D and angle bisector of $\angle A$ meets BC at E , then

$$\angle ABC + \angle ACD = 2 \angle AEC.$$



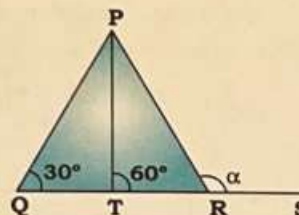
Ex. The side QR of $\triangle PQR$ is produced to a point S . The bisector of $\angle P$ meets side QR at T . If $\angle PQR = 30^\circ$ and $\angle PTR = 60^\circ$, find $\angle PRS$.

HINTS

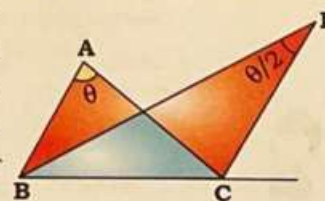
$$\angle PQR + \angle PRS = 2 \times \angle PTR$$

$$\Rightarrow 30^\circ + \alpha = 2 \times 60^\circ$$

$$\Rightarrow \alpha = 120^\circ - 30^\circ = 90^\circ$$



In a triangle, the angle formed between internal bisector of one base angle and external bisector of the other base angle is half of the remaining vertex angle.



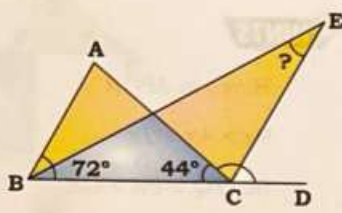
$$\text{According to this property, } \angle BEC = \frac{\angle A}{2}.$$

Ex. In $\triangle ABC$, $\angle B = 72^\circ$ and $\angle C = 44^\circ$. Side BC is produced to D. Then bisectors of $\angle B$ and $\angle ACD$ meet at E. What is the measure of $\angle BEC$?

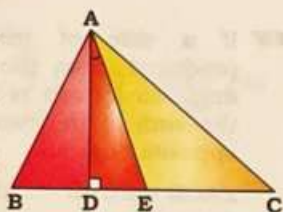
HINTS We know,

$$\begin{aligned}\angle A + \angle B + \angle C &= 180^\circ \\ \Rightarrow \angle A &= 180^\circ - 72^\circ - 44^\circ \\ \Rightarrow \angle A &= 64^\circ\end{aligned}$$

$$\therefore \angle BEC = \frac{\angle A}{2} = \frac{64^\circ}{2} = 32^\circ$$



Ex. The angle between perpendicular drawn from a vertex to opposite side and angle bisector of that vertex angle is half of difference between other two remaining vertex angles.



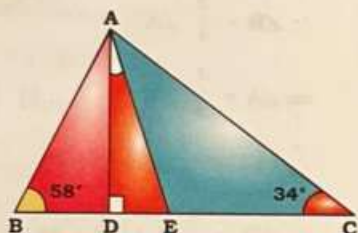
If $AD \perp BC$ and AE is angle bisector of $\angle A$, then

$$\angle DAE = \frac{|\angle B - \angle C|}{2}$$

Ex. In $\triangle ABC$, AD is perpendicular to BC and AE is the bisector of $\angle BAC$. If $\angle ABC = 58^\circ$ and $\angle ACB = 34^\circ$, then find the measure of $\angle DAE$.

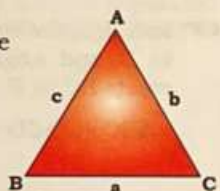
HINTS

$$\begin{aligned}\angle DAE &= \frac{\angle B - \angle C}{2} \\ &= \frac{58^\circ - 34^\circ}{2} = \frac{24^\circ}{2} \\ &= 12^\circ\end{aligned}$$



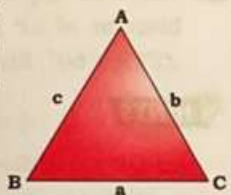
(i) Sum of any two sides of a triangle is always greater than the third side.

$$\begin{aligned}a + b &> c, \quad b + c > a \\ c + a &> b\end{aligned}$$



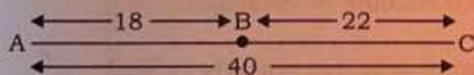
(ii) The difference of the length of any two sides of a triangle is always smaller than the third side.

$$\begin{aligned}|a - b| &< c, \quad |b - c| < a \\ |c - a| &< b\end{aligned}$$



1. If one side of a triangle is greater than the sum of the other two sides then a triangle can't be formed.

2. When the length of one side is equal to sum of the length of other two sides then the points are collinear & triangle is not formed (just a straight line back and forth). i.e., if $a + b = c$ then point A, B and C are collinear.



Ex. Three sides of a triangle are 5 cm, 9 cm, and x cm. The minimum integral value of x is.

HINTS

Value of x lies between, $4 < x < 14$

Thus, Minimum integral value of x is 5.



Ex. If the sides of a triangle are 7, 12 and x , and x is an integer, then find the number of possible values of x .

HINTS $12 + 7 > x > 12 - 7$

$$\Rightarrow 19 > x > 5$$

$$x = (19 - 5) - 1 = 13$$

Alternatively

No. of possible values of $x = 2 \times \text{smallest side} - 1$

$$= 2 \times 7 - 1 = 13$$



"Number of Possible value of x
= $2 \times \text{smallest side} - 1$ "

Area of Triangle

(A) Area of $\triangle ABC = \frac{1}{2} \times \text{base} \times \text{height}$

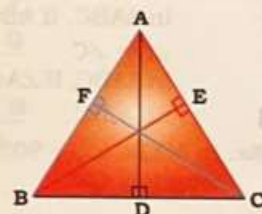


To calculate the area of $\triangle ABC$, we take any of the sides as base and the corresponding perpendicular from the vertex to the base as the height.

In $\triangle ABC$,
 $AD \perp BC$, $BE \perp AC$ and
 $CF \perp AB$.

$$\text{ar}(\triangle ABC) = \frac{1}{2} \times BC \times AD$$

$$= \frac{1}{2} \times AC \times BE = \frac{1}{2} \times AB \times CF$$



(i) If the height of the two triangles is equal, the ratio of their areas is proportional to the ratio of their bases.

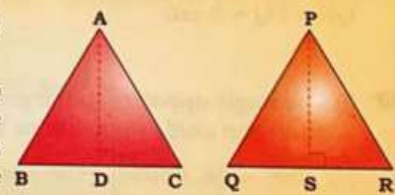
• If $AD = PS$, then

$$\begin{aligned}\text{Ar}(\triangle ABC) : \text{Ar}(\triangle PQR) \\ = BC : QR\end{aligned}$$

(ii) If the bases of the two triangles is equal, the ratio of their areas is proportional to the ratio of their height.

• If $BC = QR$, then

$$\text{Ar}(\triangle ABC) : \text{Ar}(\triangle PQR) = AD : PS$$



(B) Area of scalene triangle

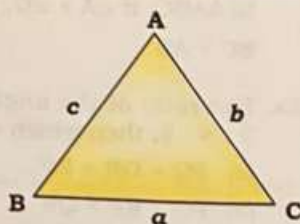
Area of $\triangle ABC$

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

(Heron's formula)

Where, Semi Perimeter (s)

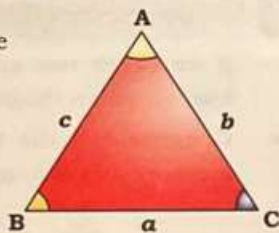
$$= \frac{a+b+c}{2}$$



- (C) When two sides and the angle between these two sides are given

$$\text{Area of } \triangle ABC = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} bc \sin A = \frac{1}{2} ac \sin B$$



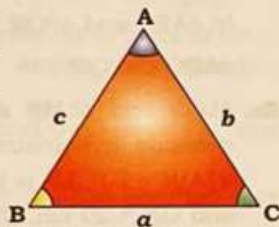
Sine & Cosine Rule

(a) Sine Rule

The ratio of side and sine of opposite angle of a triangle is equal to twice of circum radius.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R;$$

Where, R is the circumradius of triangle.



(b) Cosine Rule

If two sides and the angle between those sides are given, then we can find the opposite side by Cosine Rule.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Ex. In $\triangle ABC$, $\angle B = 30^\circ$ and $\angle C = 45^\circ$. If $BC = 50\text{cm}$, then find the length of AB .

HINTS

$$\Rightarrow \frac{50}{\sin 105^\circ} = \frac{x}{\sin 45^\circ}$$

$$\Rightarrow \frac{50}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \frac{x}{\frac{1}{\sqrt{2}}}$$

$$\left(\because \sin 105^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} \right)$$

$$\Rightarrow x = \frac{100}{(\sqrt{3}+1)} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 50(\sqrt{3}-1)$$

Ex. In $\triangle ABC$, $AB = 12\text{cm}$ and $AC = 10\text{cm}$, and $\angle BAC = 60^\circ$. What is the length of side BC ?

HINTS

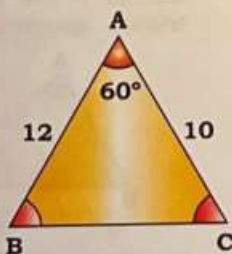
$$\cos A = \frac{AB^2 + AC^2 - BC^2}{2 \times AB \times AC}$$

$$\Rightarrow \cos 60^\circ = \frac{(12)^2 + (10)^2 - BC^2}{2 \times 12 \times 10}$$

$$\Rightarrow \frac{1}{2} = \frac{144 + 100 - BC^2}{2 \times 120}$$

$$\Rightarrow 120 = 244 - BC^2$$

$$\Rightarrow BC^2 = 124 \Rightarrow BC = 2\sqrt{31} \text{ cm}$$



Congruency & Similarity in Triangles

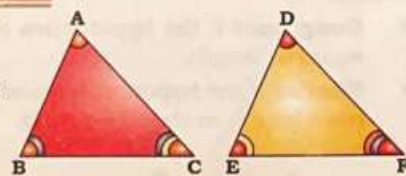
Congruency in Triangles

- Two triangles are Congruent, if

- (i) Their corresponding angles are equal

$$\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$$

- (ii) Their corresponding sides are equal.
 $AB = DE, BC = EF, CA = FD$



Similarity in Triangles

- Two triangles are similar, if

- (i) Their corresponding angles are equal

$$\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$$

- (ii) Their corresponding sides are in the same ratio (or proportion).

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$



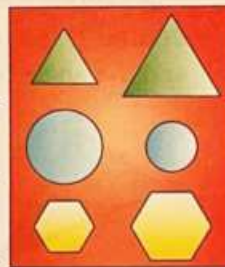
All congruent figures are similar, but not all similar figures are congruent.

Similarity in Regular Figures

Any two figures could be tested to check for similarity and congruence. In the case of regular figures, this is easiest – any two regular figures with the same number of sides will be similar to each other.

For example if we take two regular hexagons, or two circles, or two equilateral triangles, or two squares, or two regular pentagons, each pair of figures will be similar without any further checking required.

The figure may not always look similar-one should test to make sure.

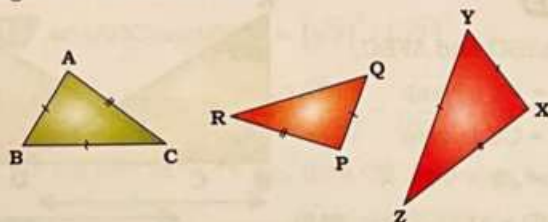


Conditions for Congruency and Similarity of Triangles

SSS (Side-Side-Side) Test

If we check the three sides of two triangles, then the triangles are

- Congruent** if three pairs of sides of the two triangles are equal in length.
- Similar** if the corresponding sides of two triangles have lengths in the same ratio.



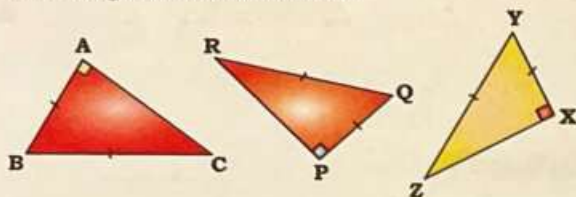
In $\triangle ABC$ and $\triangle PQR$: $AB = PQ, BC = QR, AC = PR$
 $\therefore \triangle ABC \cong \triangle PQR$

In $\triangle PQR$ and $\triangle XYZ$: $\frac{PQ}{XY} = \frac{QR}{YZ} = \frac{PR}{XZ}$
 $\therefore \triangle PQR \sim \triangle XYZ$

Hypotenuse Side Test

If we check the sides of two right-angled triangles, then the triangles are

- **Congruent** if the hypotenues and one shorter sides are equal in length.
- **Similar** if the hypotenues and one pair of shorter sides have length in the same ratio.



In $\triangle ABC$ and $\triangle PQR$: $AB = PQ$, $BC = QR$

$$\therefore \triangle ABC \cong \triangle PQR$$

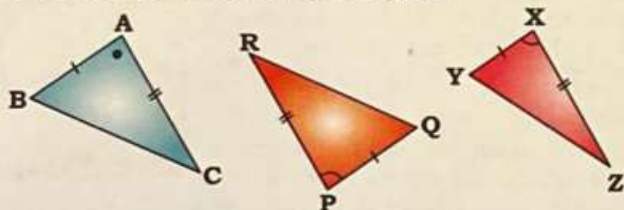
In $\triangle PQR$ and $\triangle XYZ$: $\frac{PQ}{XY} = \frac{QR}{YZ}$

$$\therefore \triangle PQR \sim \triangle XYZ$$

SAS (Side-Angle-Side) Test

If we check two side and its included angle of two triangles then the triangle are

- **Congruent** if the two sides are equal in length and the angle between equal sides is also equal.
- **Similar** if the two sides have lengths in the same ratio and the angle between them is equal.



In $\triangle ABC$ and $\triangle PQR$: $AB = PQ$, $AC = PR$, $\angle A = \angle P$

$$\therefore \triangle ABC \cong \triangle PQR$$

In $\triangle PQR$ and $\triangle XYZ$: $\frac{PQ}{XY} = \frac{QR}{YZ}$, $\angle P = \angle X$

$$\therefore \triangle PQR \sim \triangle XYZ$$

Ex. In $\triangle ABD$ and $\triangle FEC$, $\angle BAD = 60^\circ$, $l(BD) = l(EC)$, $\angle ABD = \angle FEC = 90^\circ$, and $l(AB) = l(FE)$. Find the ratio of $\angle BAD$ and $\angle FCE$.

HINTS

In $\triangle ABD$ and $\triangle FEC$,

$AB = FE$ (given)

$BD = CE$ (given)

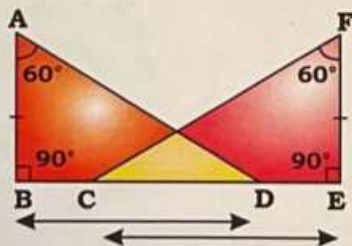
$\angle B = \angle E = 90^\circ$

$\triangle ABD \cong \triangle FEC$ (from SAS)

$\Rightarrow \angle F = 60^\circ$

$\therefore \angle FCE = 180^\circ - 90^\circ - 60^\circ = 30^\circ$

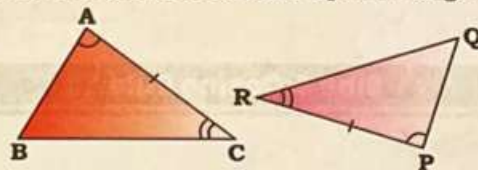
$$\therefore \frac{\angle BAD}{\angle FCE} = \frac{60^\circ}{30^\circ} = \frac{2}{1} = 2 : 1$$



ASA (Angle-Side-Angle) Test

If we check two angles and the included side of two triangles then the triangle are

- **Congruent** if the two pairs of angle have the same measure and only one side are equal in length.



In $\triangle ABC$ and $\triangle PQR$: $BC = QR$, $\angle A = \angle P$, $\angle C = \angle R$

$$\triangle ABC \cong \triangle PQR$$

Ex. $\triangle LON$ and $\triangle LMN$ are two right-angled triangles with common hypotenuse LN such that $\angle LON = 90^\circ$ and $\angle LMN = 90^\circ$. LN is the bisector of $\angle OLM$. If $LN = 29$ cm and $ON = 20$ cm, then what is the perimeter (in cm) of $\triangle LMN$?

HINTS

In $\triangle LON$ and $\triangle LMN$,

$\angle LON = \angle LMN = 90^\circ$

$LN = LN$ (common side)

$\angle OLN = \angle MLN$ (angle bisector)

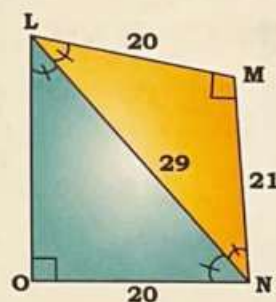
$\therefore \triangle LON \cong \triangle LMN$ (By ASA)

In $\triangle LMN$,

$$MN^2 = 29^2 - 20^2 \Rightarrow MN = 21$$

$\therefore$ Perimeter of $\triangle LMN = 29 + 20 + 21$

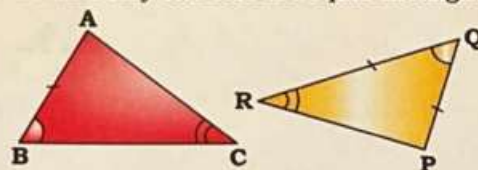
$$= 70 \text{ cm}$$



AAS (Angle-Angle-Side) Test

If we check two angles and a corresponding non included side of two triangles then the triangle are

- **Congruent** if the two pairs of angles have the same measure and only one side are equal in length.



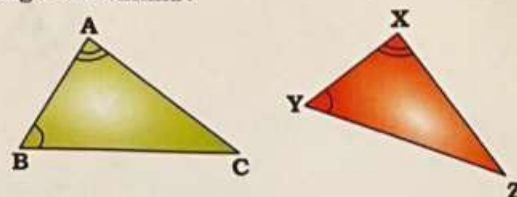
In $\triangle ABC$ and $\triangle PQR$: $AB = PQ$, $\angle B = \angle Q$, $\angle C = \angle R$

$$\triangle ABC \cong \triangle PQR$$

AA Test

If we check the angles of two triangles then the triangles are

- Similar if two angles of triangle are equal then both triangle are similar.

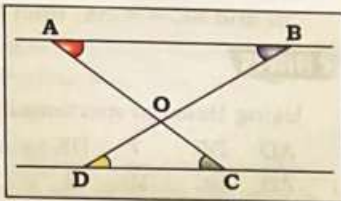


In $\triangle ABC$ and $\triangle PQR$: $\angle A = \angle X$, $\angle B = \angle Y$

$$\triangle ABC \sim \triangle XYZ$$

Spotting Similarity & Congruency

- Identifying similarity (and congruency) is crucial in geometry, especially for visualizing problems.
- A key to recognizing similarity is spotting equal angles.



Case 01: Parallel Lines

- When you see two parallel lines intersected by transversals, immediately look for similar triangles.
- This is because parallel lines create many pairs of equal angles (e.g., alternate interior angles, corresponding angles).

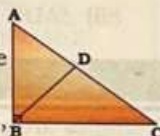
Given parallel lines AB and CD, and transversals AC and BD intersecting at O:

- Triangles $\triangle AOB$ and $\triangle COD$ are similar because they have two pairs of equal angles:
 - $\angle OAB = \angle OCD$ (Alternate interior angles)
 - $\angle OBA = \angle ODC$ (Alternate interior angles)
 - $\angle AOB = \angle COD$ (Vertically opposite angles)

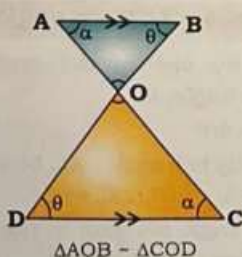
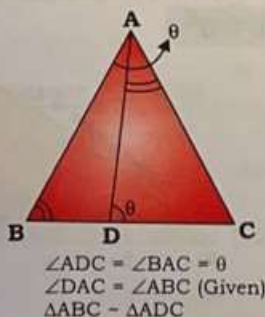
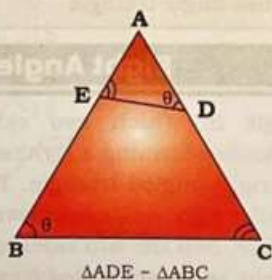
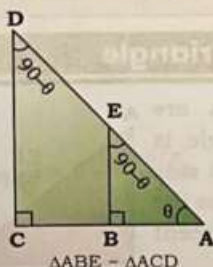
Case 02: Altitude to the Hypotenuse in a Right Triangle:

When you draw an altitude from the right angle vertex to the hypotenuse of a right-angled triangle, it creates three similar triangles:

- The original right triangle.
- The two smaller triangles formed by the altitude.
- Example (using $\triangle ABC$ right-angled at B, with BD as altitude to AC):
 - $\triangle ABC \sim \triangle ADB$ (They share $\angle A$, and both have a right angle).
 - $\triangle ABC \sim \triangle BDC$ (They share $\angle C$, and both have a right angle).
 - Therefore, all three triangles are similar to each other: $\triangle ABC \sim \triangle ADB \sim \triangle BDC$.



Some Similar Figure



Ex. In $\triangle ABC$, $\angle B = 87^\circ$ and $\angle C = 60^\circ$. Points D and E are on the sides AB and AC, respectively, such that $\angle DEC = 93^\circ$ and DAE is

HINTS

$$\angle DEA = \angle ABC$$

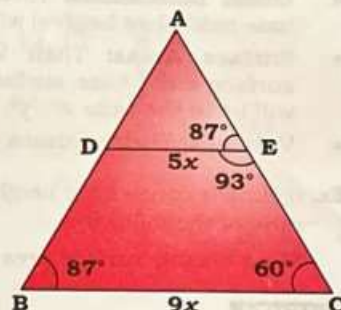
$$\angle A = \text{Common}$$

$$\Rightarrow \triangle AED \sim \triangle ABC$$

$$\Rightarrow \frac{AE}{AB} = \frac{ED}{BC} = \frac{AD}{AC}$$

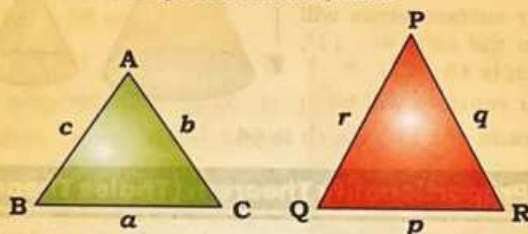
$$\Rightarrow \frac{AE}{14.4} = \frac{5}{9}$$

$$\Rightarrow AE = 8 \text{ cm}$$



Properties in Similar triangles

If $\triangle ABC$ and $\triangle PQR$ are similar, then



$$(i) \frac{a}{p} = \frac{b}{q} = \frac{c}{r}$$

(ii) Ratio of corresponding sides

- = Ratio of perimeter
- = Ratio of semi-perimeter(s)
- = Ratio of corresponding medians
- = Ratio of inradius
- = Ratio of circumradius

(iii) Ratio of area

- = Ratio of square of corresponding sides
- = Ratio of square of perimeter
- = Ratio of square of semi-perimeter
- = Ratio of square of corresponding medians
- = Ratio of square of inradius
- = Ratio of square of circumradius

Ex. If the ratio of corresponding sides of two similar triangles is $\sqrt{3} : \sqrt{2}$, then what is the ratio of the area of the two triangles?

HINTS

$$\text{ar}(\triangle ABC) : \text{ar}(\triangle PQR) = (\sqrt{3})^2 : (\sqrt{2})^2 = 3 : 2$$

Ex. Let $\triangle ABC \sim \triangle QPR$ and $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle QPR)} = \frac{64}{169}$. If $AB = 10 \text{ cm}$,

$BC = 7 \text{ cm}$ and $AC = 16 \text{ cm}$, then QR (in cm) is equal to:

HINTS

Here, $\triangle ABC \sim \triangle QPR$ & $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle QPR)} = \frac{64}{169}$

$$\Rightarrow \frac{AC}{QR} = \sqrt{\frac{64}{169}} \Rightarrow \frac{16}{QR} = \frac{8}{13}$$

$$\Rightarrow QR = \frac{13 \times 16}{8} = 26 \text{ cm}$$

Similarity of Cones

If two cones are similar and their heights (a linear dimension) are in the ratio $x : y$.

- **Linear Dimensions:** Their other linear dimensions (like base radii, slant heights) will also be in the same ratio $x : y$.
- **Surface Areas:** Their various surface areas (total surface area, base surface area, curved surface area) will be in the ratio $x^2 : y^2$.
- **Volumes:** Their volumes will be in the ratio $x^3 : y^3$.

Ex. If similar cones have heights in the ratio $4 : 11$. Find the ratio of the following:

Slant height, surface area and volume

HINTS

Their base radii and slant heights will also be in the ratio $4 : 11$.

Their surface areas will be in the ratio $4^2 : 11^2$, which is $16 : 121$.

Their volumes will be in the ratio $4^3 : 11^3$, which is $64 : 1331$.



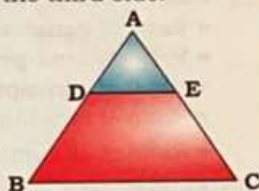
Basic Proportionality Theorem (Thales Theorem)

- A line drawn parallel to one side of a triangle divides other two sides in the same ratio.
- If a line divides any two sides of a triangle in the same ratio, the line must be parallel to the third side.

In $\triangle ABC$, If $DE \parallel BC$, then

$$\frac{AD}{DB} = \frac{AE}{EC}$$

'OR' If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$



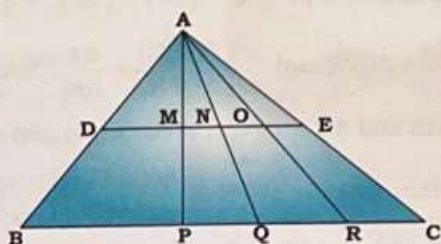
- Some of the results derived from this theorem, we will use, are as follows:

$$(i) \frac{AD}{BD} = \frac{AE}{EC} \quad (ii) \frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

$$(iii) \triangle ADE \sim \triangle ABC$$

$$(iv) \frac{Ar(\triangle ADE)}{Ar(\triangle ABC)} = \left(\frac{AD}{AB}\right)^2 = \left(\frac{AE}{AC}\right)^2 = \left(\frac{DE}{BC}\right)^2$$

- A line drawn parallel to one side of a triangle divides the median, the angle bisector and the altitude of triangle in the same ratio it divides the other two sides of the triangle.



In $\triangle ABC$, AP, AQ and AR are the median, the angle bi-sector and altitude respectively and $DE \parallel BC$, then

$$\frac{AD}{DB} = \frac{AE}{EC} = \frac{AM}{MP} = \frac{AN}{NQ} = \frac{AO}{OR}$$

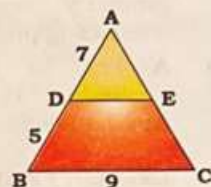
Ex. In $\triangle ABC$, D and E are the points on sides AB and AC, respectively such that $\angle ADE = \angle B$. If $AD = 7$ cm $BD = 5$ cm and $BC = 9$ cm, then DE (in cm) is equal to ____

HINTS

Using Basic proportionality theorem,

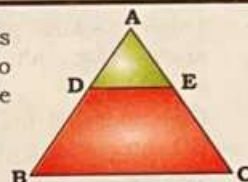
$$\frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{7}{12} = \frac{DE}{9}$$

$$\Rightarrow DE = \frac{63}{12} = 5.25 \text{ cm}$$



Mid-Point Theorem

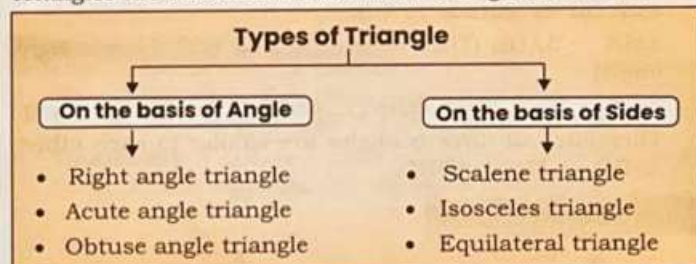
The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the magnitude of third side and vice-versa.



- If D and E are mid-points of AB and AC, respectively, then $DE \parallel BC$ and $DE = \frac{BC}{2}$
- $DE \parallel BC$ and $DE = \frac{BC}{2}$, then D and E are the mid-points of AB and AC respectively. In this case
 - (i) $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC} = \frac{1}{2}$
 - (ii) $\frac{AD}{DB} = \frac{AE}{EC} = 1$
 - (iii) $\triangle ADE \sim \triangle ABC$
 - (iv) $\frac{Ar(\triangle ADE)}{Ar(\triangle ABC)} = \frac{1}{4}$

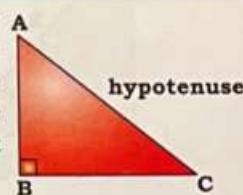
Types of Triangle

Triangles are classified on the basis of angles and sides—



Right Angle Triangle

A triangle in which two sides are perpendicular, forming a right angle, is called a right-angled triangle. The side opposite the right angle is known as the hypotenuse, and the two sides adjacent to the right angle are called legs.



Properties of a Right angle Triangle

- (i) Exactly one of the angle is right angle, i.e. $\angle B = 90^\circ$
- Exactly two angles will be acute. i.e. $\angle A < 90^\circ$, $\angle C < 90^\circ$
- One angle is equal to the sum of other two angle, i.e. $\angle B = \angle A + \angle C = 90^\circ$



- (ii) (a) **Pythagoras Theorem:** In a right triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

If a , b and c be three sides of a right-angled triangle, then according to the Pythagoras Theorem,

$$a^2 + b^2 = c^2$$

For Example: $3^2 + 4^2 = 5^2$

- (b) **Pythagorean Triplets:** A set of three integers a , b , c which satisfy Pythagoras Theorem ($a^2 + b^2 = c^2$) or are the sides of a right angle triangle is called Pythagorean triplets.

- The smallest Pythagorean triplet is (3, 4, 5).
- If (a, b, c) be a Pythagorean triplets, then $(ak, bk,$

$ck)$ or $(\frac{a}{k}, \frac{b}{k}, \frac{c}{k})$ will also be the Pythagorean triplet.

Following Pythagorean triplets are frequently used in the examinations.

3	4	5	$\times \frac{1}{2}$	1.5, 2, 2.5	$1, 1, \sqrt{2}$ $9, 40, 41$ $11, 60, 61$ $12, 35, 37$ $16, 63, 65$ $13, 84, 85$ $20, 21, 29$ $39, 80, 89$ $36, 77, 85$ $65, 72, 97$ $20, 99, 101$
			$\times 3$	9, 12, 15	
			$\times 2$	6, 8, 10	
			$\times 4$	12, 16, 20	
			$\times 5$	15, 20, 25	
5	12	13	$\times 2$	10, 24, 26	
			$\times \frac{1}{2}$	3.5, 12, 12.5	
7	24	25	$\times 2$	14, 48, 50	
			$\times 3$	21, 72, 75	
$(2n, n^2 - 1, n^2 + 1)$				$(\frac{n^2 + 1}{2}, \frac{n^2 - 1}{2}, n)$	

Tricks for Making Triplets

- (a) For even number $\rightarrow$

$$6 \rightarrow \frac{6}{2} = 3 \xrightarrow{\text{Square}} 9 \begin{cases} -1 & 8 \\ +1 & 10 \end{cases}$$

$$8 \rightarrow \frac{8}{2} = 4 \xrightarrow{\text{Square}} 16 \begin{cases} -1 & 15 \\ +1 & 17 \end{cases}$$

$$12 \rightarrow \frac{12}{2} = 6 \xrightarrow{\text{Square}} 36 \begin{cases} -1 & 35 \\ +1 & 37 \end{cases}$$

- (b) For odd number $\rightarrow$

$$3 \xrightarrow{\text{Square}} 9 \rightarrow \frac{9}{2} = 4.5 \begin{cases} -0.5 & 4 \\ +0.5 & 5 \end{cases}$$

$$5 \xrightarrow{\text{Square}} 25 \rightarrow \frac{25}{2} = 12.5 \begin{cases} -0.5 & 12 \\ +0.5 & 13 \end{cases}$$

$$7 \xrightarrow{\text{Square}} 49 \rightarrow \frac{49}{2} = 24.5 \begin{cases} -0.5 & 24 \\ +0.5 & 25 \end{cases}$$

"(20, 21, 29) and (20, 99, 101) are triplets."

- Ex.** $\triangle ABC$ is right angled at B and D is a point of BC such that $BD = 5$ cm, $AD = 13$ cm and $AC = 37$ cm, then find the length of BC in cm.

HINTS

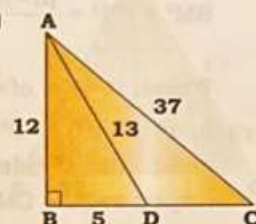
$AB = 12$ cm (By triplet of 5, 12, 13)

$$(12)^2 + (BC)^2 = (37)^2$$

$$\Rightarrow (BC)^2 = (37)^2 - (12)^2$$

$$\Rightarrow BC^2 = 25 \times 49$$

$$\Rightarrow BC = 5 \times 7 = 35$$



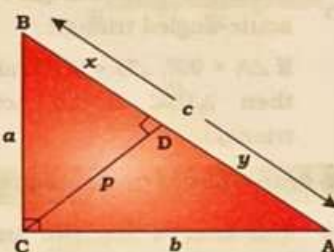
Alternatively

(5, 12, 13) and (12, 35, 37) are triplet.

With the help of triplet, $AB = 12$ cm satisfy the given condition.

$$\therefore BC = 35 \text{ cm}$$

- (iii) In a right angle $\triangle ABC$, if a perpendicular (CD) is drawn from the vertex which is right angle ($\angle C$) to the hypotenuse (AB), then,
 $\triangle ABC \sim \triangle BCD \sim \triangle ACD$



$$\begin{aligned} & \bullet a^2 = x \times c & \bullet p^2 = x \times y & \bullet b^2 = y \times c \\ & \bullet a \times b = p \times c & \bullet \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} \end{aligned}$$

- Ex.** In $\triangle ABC$, $\angle A = 90^\circ$, $AD \perp BC$ at D. If $AB = 12$ cm and $AC = 16$ cm, then what is the length (in cm) of BD?

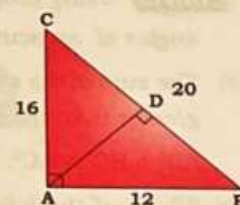
HINTS

$$AB^2 = BD \times BC$$

$$\Rightarrow 12^2 = BD \times 20$$

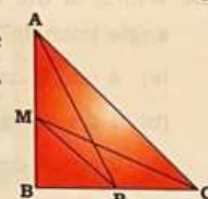
$$\Rightarrow 144 = BD \times 20$$

$$\Rightarrow BD = 7.2 \text{ cm}$$



- (iv) If P and M are the two points on the sides BC and AB respectively of $\triangle ABC$, right-angled at B, then

$$AP^2 + MC^2 = AC^2 + MP^2$$



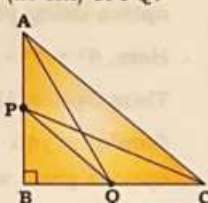
- Ex.** Points P and Q are on the sides AB and BC respectively of a $\triangle ABC$, right angled at B. If $AQ = 11$ cm, $PC = 8$ cm, and $AC = 13$ cm, then find the length (in cm) of PQ.

HINTS

$$AQ^2 + PC^2 = AC^2 + PQ^2$$

$$\Rightarrow 121 + 64 = 169 + PQ^2$$

$$\Rightarrow 16 = PQ^2 \Rightarrow PQ = 4$$



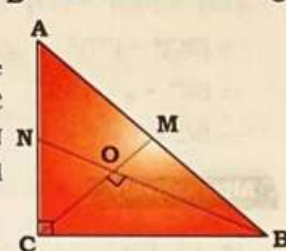
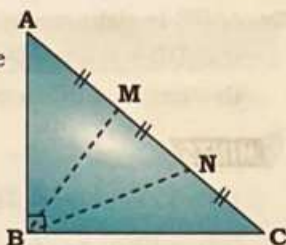
- (v) $\triangle ABC$ is a right angle triangle, right angle at B. If M, N are points on AC such that $AM = MN = NC$ then

$$BM^2 + BN^2 = \frac{(n-1)(2n-1)AC^2}{6n}$$

Where, $n = \text{No. of equal parts}$

- (vi) In $\triangle ABC$, $\angle C = 90^\circ$, M and N are mid-points of sides AB and AC respectively. CM and BN intersect each other at O and

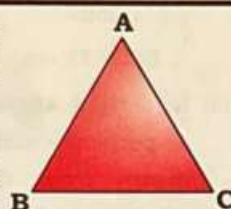
$$\angle BOC \text{ is } 90^\circ \text{ then } BN = \frac{BC\sqrt{6}}{2}$$



Acute Angle Triangle

If each of the angles of a triangle is acute, then the triangle is called an acute-angled triangle.

If $\angle A < 90^\circ$, $\angle B < 90^\circ$ and $\angle C < 90^\circ$, then $\triangle ABC$ is an acute-angled triangle.



Properties of an acute angle triangle

- (i) The sum of any two angles of the triangle is always greater than the third angle.

$$\angle A + \angle B > \angle C, \angle B + \angle C > \angle A, \angle C + \angle A > \angle B$$

Ex. Which one of the following cannot be the ratio of angles in acute angle triangle?

- (a) 2 : 5 : 8 (b) 4 : 1 : 4
(c) 2 : 3 : 4 (d) 1 : 1 : 1

HINTS Using property (i), Option (a) can't be the ratio of angles of an acute triangle.

- (ii) The sum of the squares of any two sides of the triangle is greater than the square of the third side.

$$AB^2 + BC^2 > AC^2, BC^2 + AC^2 > AB^2, AC^2 + AB^2 > BC^2$$

Ex. Which of the following can be the 3 sides of an acute angle triangle?

- (a) 4 cm, 8 cm, 7 cm
(b) 1 cm, 2 cm, 3 cm
(c) 2 cm, 4 cm, 3 cm
(d) 9 cm, 12 cm, 15 cm

HINTS In such type of question you have to check each option using property (ii).

$$\text{Here, } 4^2 + 8^2 > 7^2, 8^2 + 7^2 > 4^2 \text{ \& } 7^2 + 4^2 > 8^2$$

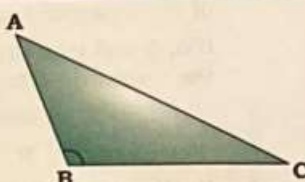
Thus Option (A) is right answer.

Similarly you can check other options but all other options given will not satisfy the property.

Obtuse Angle Triangle

If one of the angle of a triangle is obtuse, then the triangle is called an obtuse-angled triangle.

$\triangle ABC$ is obtuse angle triangle.



Properties of an obtuse-angled triangle

- (i) Exactly one of the angle of the triangle is obtuse and other two angles are acute.

$$\angle B > 90^\circ \text{ and } 0^\circ < \angle A, \angle C < 90^\circ$$

- (ii) The sum of the two acute angles of the triangle is less than the obtuse angle.

$$(\angle A + \angle C) < \angle B$$

- (iii) The sum of the square of two smaller sides is less than the square of the third side (the largest side).

$$AB^2 + BC^2 < AC^2$$

Ex. In an obtuse angle $\triangle PQR$, $\angle Q$ is obtuse angle if side $PQ = 11\text{cm}$, $QR = 15\text{cm}$ then find minimum possible integer length of side PR ?

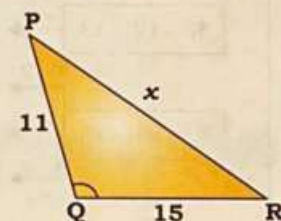
HINTS

$$11^2 + 15^2 < x^2$$

$$\Rightarrow 121 + 225 < x^2$$

$$\Rightarrow 346 < x^2$$

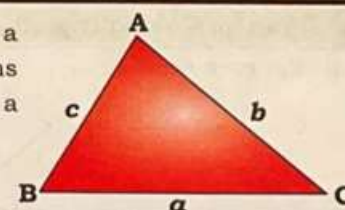
$$\Rightarrow x_{\min} = 18.60 \sim 19 \text{ cm}$$



Scalene Triangle

If all of the three sides of a triangle are of different lengths then the triangles is called a scalene triangle.

$$AB \neq BC \neq CA \Rightarrow a \neq b \neq c$$



Properties of a scalene triangle

- (i) No two sides are equal in length.

$$AB \neq BC \neq CA \text{ or } a \neq b \neq c$$

- (ii) No two angles are equal, i.e., $\angle A \neq \angle B \neq \angle C$

- (iii) Triangle may be one of the acute angled, right angled or obtuse angled.

- (iv) **Perimeter, Semi-perimeter and Area of scalene triangle**

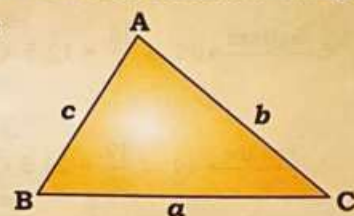
$$\bullet \text{ Perimeter} = a + b + c$$

$$\bullet \text{ Semi-perimeter(s)}$$

$$= \frac{a+b+c}{2}$$

$$\bullet \text{ Area of } \triangle ABC$$

$$= \sqrt{s(s-a)(s-b)(s-c)}$$



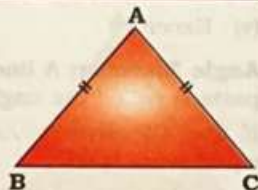
Ex. The sides of the scalene triangle ABC are in the ratio 3 : 5 : 6 and the semi-perimeter is 42 cm, what is the difference of the largest and the smallest sides of the triangle?

HINTS Semi-perimeter = $\frac{3x+5x+6x}{2}$
 $\Rightarrow 42 = 7x \Rightarrow x = 6$
 Required Difference = $6x - 3x = 3 \times 6 = 18$ cm

Isosceles Triangle

If two sides of a triangle are equal in length, then the triangle is called an isosceles triangle.

If $AB = AC \neq BC$, then $\triangle ABC$ is an isosceles triangle.



Properties of an isosceles triangle

(i) The length of the two sides are equal, but the length of third side can be smaller or larger than the equal sides.

$AB = AC$ & $BC > AB$ or $BC < AB$

(ii) Two angles opposite to the equal sides are equal.

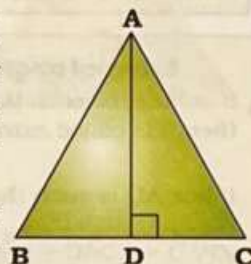
$\angle B = \angle C$ (opposite to the side AC and AB)

(iii) In an isosceles $\triangle ABC$, the median, the angle bisector, perpendicular bisector and the altitude of unequal side is the same and it divides $\triangle ABC$ in two congruent (equal) right-angled triangle.

In $\triangle ABC$, AD is the median, the angle bisector, perpendicular bisector and the altitude.

$AD \perp BC$ and $BD = DC$, then

$\triangle ABD \cong \triangle ACD$



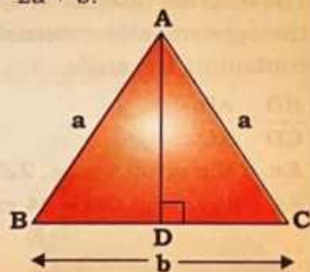
(iv) **Perimeter, semi-perimeter and Area**

• Perimeter = $AB + BC + CA = 2a + b$.

• Semi perimeter
 $= \frac{AB + BC + CA}{2} = a + \frac{b}{2}$

• $AD = \frac{\sqrt{4a^2 - b^2}}{2}$

• Area of $\triangle ABC = \frac{b}{4} \sqrt{4a^2 - b^2}$



In isosceles triangle ABC, $\triangle ABD$ and $\triangle ACD$ are the two congruent right-angled triangle. Use Pythagorean triplets in these right angled triangle.

Ex. The perimeter of an isosceles triangle is 50 cm. If the base is 18 cm, then find the length of equal sides.

HINTS Perimeter, $2a + b = 50$
 $\Rightarrow 2a + 18 = 50$
 $\Rightarrow a = 16$

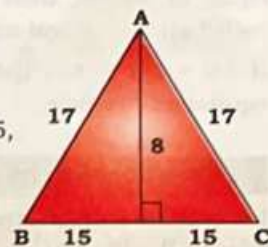
Ex. The length of perpendicular dropped on the base of an isosceles triangle is 8 cm. If its perimeter is 64 cm, what is the area of this triangle?

HINTS

Check $30 + 34 = 64$

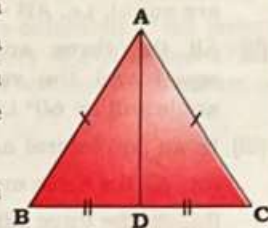
[Apply concept of triplet, (8, 15, 17) to solve quickly]

Area = $\frac{1}{2} \times 30 \times 8 = 120$ cm²



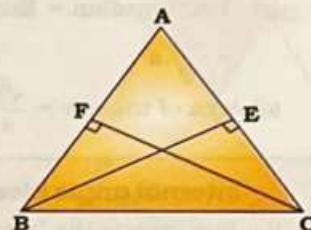
(v) If a line (which joins the common vertex of two equal sides of a triangle) bisects the base then that line is perpendicular to the base and vice versa.

If $AB = AC$ and $BD = DC$, then
 $\angle ADB = \angle ADC = 90^\circ$

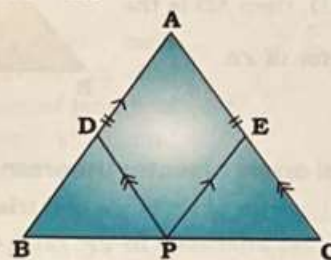


(vi) • If $AB = AC$ and $BE \perp AC$ and $CF \perp AB$, then
 $BE = CF$

• If $AB = AC$, E and F are the mid-points, then
 $BE = CF$



(vii) ABC is an isosceles triangle ($AB = AC$). If P is a point on the side BC and $DP \parallel AC$ and $EP \parallel AB$ then
 $DP + PE = AB = AC$



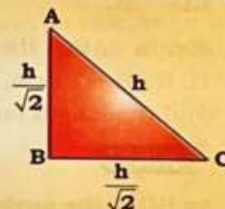
Isosceles Right Angle Triangle

• Ratio of sides

$\Rightarrow AB : BC : AC = 1 : 1 : \sqrt{2}$

• Area = $\frac{\text{Hypotenuse}^2}{4}$

• Perimeter = $h(\sqrt{2} + 1)$



Ex. If the hypotenuse of the isosceles right angle triangle equals the side of a square whose area is 256 cm². Find the area of triangle.

HINTS Side of square = $\sqrt{256} = 16$.

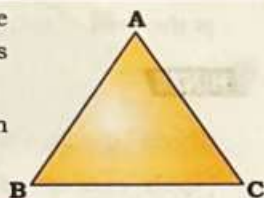
$\therefore$ side of square = $h = 16$

Area = $\frac{h^2}{4} = \frac{16^2}{4} = \frac{256}{4} = 64$ cm².

Equilateral Triangle

If all the three sides of a triangle are equal in length, then the triangle is called an equilateral triangle.

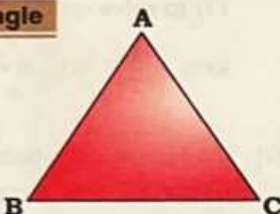
If $AB = BC = AC$, then $\triangle ABC$ is an equilateral triangle.



Properties of an equilateral triangle

Let $\triangle ABC$ is an equilateral triangle in which $AB = BC = AC$

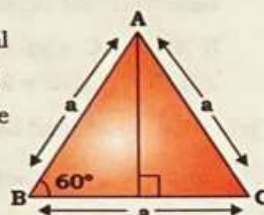
- (i) The length of all the three sides are equal, i.e. $AB = BC = AC$
- (ii) All the three angles will be equal and the value of each angle will be 60° i.e. $\angle A = \angle B = \angle C$
- (iii) In an equilateral $\triangle ABC$,



- (a) All the three medians are equal
- (b) All the three altitudes are equal
- (c) Each median = Each altitude

$$= \frac{\sqrt{3}}{2} a$$

$$(d) \text{ Area of triangle} = \frac{\sqrt{3}}{4} a^2$$



- (iv) In equilateral $\triangle ABC$, incentre, circumcentre, orthocentre and centroid are at the same point.
- (v) If altitudes of a triangle are equal, the triangle is an equilateral triangle.

Centres of Triangle

We will study 5 types of centres of a triangle

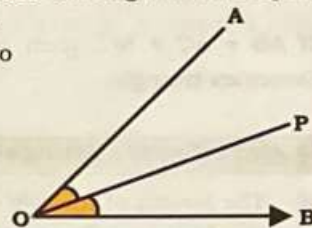
- (i) Incentre
- (ii) Circumcenter
- (iii) Orthocentre
- (iv) Centroid
- (v) Excentre

Angle Bisector: A line which divides an angle in two equal parts is called the angle bisector.

If line OP divides $\angle AOB$ in two equal parts then

$$\angle AOP = \angle BOP = \frac{\angle AOB}{2}, \text{ then}$$

OP is an angle bisector.



Angle bisector of an angle of a triangle

Interior or Internal

Exterior or External

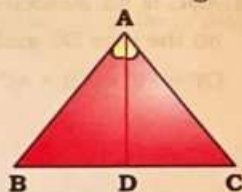
Internal angle bisector of a triangle

If a line bisects the Internal angle of a triangle, then it is called internal angle bisector of a triangle.

If line AD is such that

$$\angle BAD = \angle CAD, \text{ then AD is the}$$

interior bisector of $\angle A$.



Internal angle bisector theorem

The Internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle, i.e.

$$\frac{BD}{CD} = \frac{AB}{AC}$$

Ex. In $\triangle ABC$, the bisector of $\angle B$ meets AC at point D. If $AB = 12$ cm, $BC = 18$ cm and $AC = 15$ cm, then find the length (in cm) of AD.

HINTS

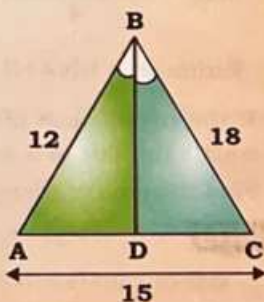
As BD is the angle bisector of $\angle ABC$

$$\Rightarrow \frac{AB}{BC} = \frac{AD}{DC} \Rightarrow \frac{12}{18} = \frac{AD}{DC}$$

$$\Rightarrow AD : DC = 2 : 3$$

Also, $AC = 15$ cm

$$\therefore AD = \frac{15}{(2+3)} \times 2 = 6 \text{ cm}$$



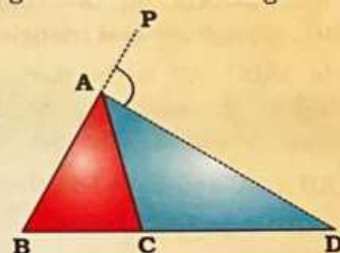
External angle bisector of a triangle

If a line bisects the exterior angle of a triangle, then it is called external angle bisector of a triangle.

If line AD is such that

$$\angle PAD = \angle DAC = \left(\frac{\pi - A}{2} \right),$$

then AD is the external bisector of $\angle A$.

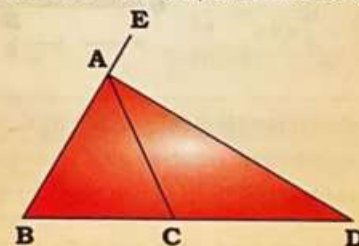


External angle bisector theorem

The external bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle

$$\frac{BD}{CD} = \frac{AB}{AC}$$

Ex. If the given figure, $2\angle EAD = \angle EAC$, $BC = 40$ cm, $BA = 8$ cm and $CD = 24$ cm, then AC is equal to



HINTS

Since AD is angle bisector

$$\frac{AB}{AC} = \frac{BD}{CD} \Rightarrow \frac{8}{AC} = \frac{64}{24} \Rightarrow AC = 3 \text{ cm}$$